How Does Geometric Bias Shape the Denoising Trajectory in Flow-Based Protein Structure Generation?

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Proteina: Geometry via Additive Pair Bias

Flow matching: SDE from noise to backbone $\quad t = 0 \xrightarrow{\text{100 SDE steps}} t = 1$

Every attention layer computes: $\quad A = \mathrm{softmax}(C + B)$

- $C = QK^\top/\sqrt{d}$, the learned content score
- B = geometric pair bias, constructed from binned pairwise Cα distances and sequence separation. Recomputed every denoising step from the current noisy structure $\mathbf{x}_t$: uninformative at t=0 (noise), encodes near-final backbone at t=1.

Two questions

When does B become causally essential during generation?
Where in the network does it act?

Methods: Four Lenses, Four Model Scales

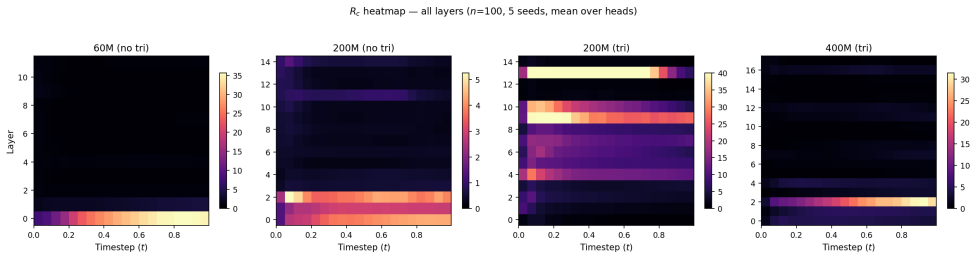
Analysis lenses:

1. **Descriptive** — how large is B vs C?
2. **Causal** — when/where must B be present?
3. **Functional** — which encodes contacts better?
4. **Representational** — where does 3D structure emerge?

Four model variants:

Model	Layers	Params	Triangle
60M no-tri	12	60M	No
200M no-tri	15	200M	No
200M tri	15	200M	Yes
400M tri	18	400M	Yes

Descriptive: Geometry Dominance in Peak Layers

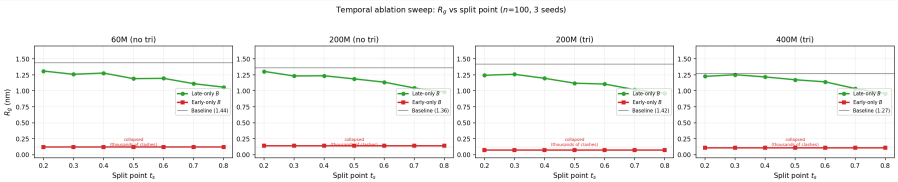


Every model concentrates geometry dominance in a few peak layers, but the location is model-specific.

But: descriptive magnitude $\neq$ causal importance — the peak-R_c layer is *not* the causally critical layer.

Causal: The Late Phase Matters; the Early Phase Does Not

	60M no-tri		200M no-tri		200M tri		400M tri	
	R_g	CI	R_g	CI	R_g	CI	R_g	CI
Early-only B ($t \leq 0.5$)	0.12	3532	0.14	2967	0.07	4702	0.11	4021
Late-only B ($t > 0.5$)	1.19	0	1.19	4	1.12	2	1.17	1



Causal: The Model Has No Memory of Early B

Gap experiment: B on $[0, 0.3]$, off $(0.3, 0.7)$, restored $[0.7, 1]$

	60M no-tri		200M no-tri		200M tri		400M tri	
Condition	R_g	Cl	R_g	Cl	R_g	Cl	R_g	Cl
Baseline	1.44	0	1.36	0	1.42	0	1.27	1
Gap (on–off–on)	1.10	0	1.04	3	1.01	3	1.05	1
Late-only (t>0.7)	1.11	0	1.04	3	1.02	2	1.04	1

Gap $\approx$ Late-only within 0.02 nm across all four models. $\Rightarrow$ **Geometric guidance is memoryless:** the trajectory forgets both the presence and absence of early B.

Causal: Architecture Inverts Which Layers Are Critical

Condition	60M no-tri	200M no-tri	200M tri	400M tri
Baseline R_g	1.44	1.36	1.42	1.27
Deep-only B	0.41	0.57	0.90	0.82
Shallow-only B	0.60	0.63	0.05	0.12
Peak-R_c only	0.81	1.28	1.17	1.13

$R_g < 0.2\,\text{nm}$ = collapse (thousands of clashes) $R_g \approx 1$–$1.5\,\text{nm}$ = valid backbone

No-tri models

Shallow layers most sensitive
Deep-only $B \rightarrow$ severe degradation

Tri models

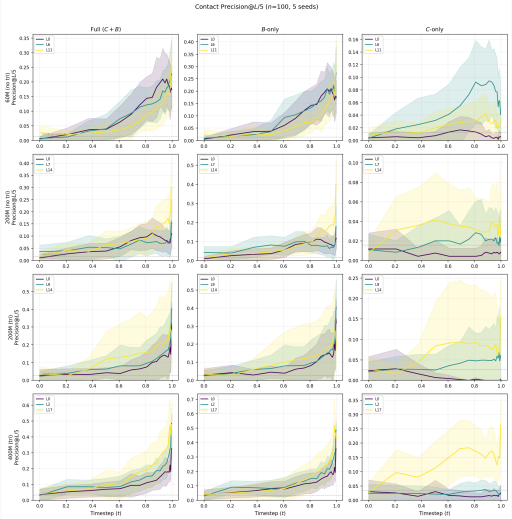
Deep layers become the bottleneck
Shallow-only $B \rightarrow$ collapse

Peak-R_c layer alone

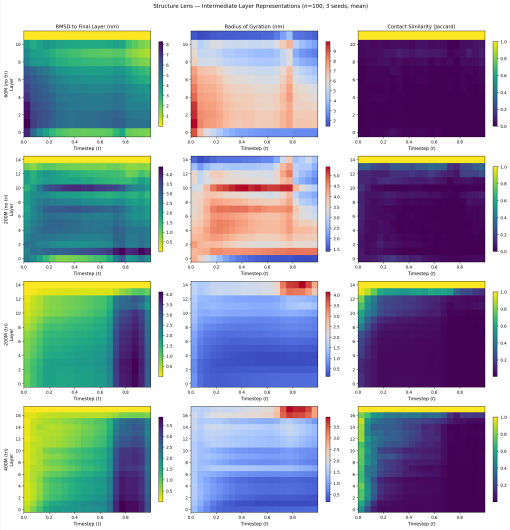
Only moderate damage $\rightarrow$ no single geometric bottleneck

Functional & Representational

B Carries the Structural Signal



Structure Emerges Only at the Output



Summary

Finding 1: Geometric guidance is memoryless

Only the *present* matters. Early B leaves no trace. Late-only B always preserves valid structures. Early-only $B \equiv$ no B.

Finding 2: Architecture determines where, not when

Triangle updates invert spatial sensitivity (shallow $\leftrightarrow$ deep), but the temporal asymmetry is identical across all four model scales.

Finding 3: Descriptive $\neq$ causal

R_c varies from 0.5 to 35 across models. Peak-R_c layer is not the critical layer. Multi-lens analysis is required to avoid misleading conclusions.

Implications & Future Work

If B is actionable only during late generation:

Late-stage geometric steering

Align intermediate representations to structure-based objectives (atomic potentials, fold classifiers) during the **final denoising steps only**.

Early steps can be left free.

Early-stage representation alignment

If early B is not used, early generation may benefit from **content-aligned objectives** (cf. REPA, Yu et al. 2025).

Connect flow models to representation alignment literature.

Code: `github.com/rs-gh/proteina-interpretability`